PRACTICE SHEET 1

1. The average of the first 7 integers in a series of 13 consecutive odd integers is 37. What is the average of the entire series?
(1) 37 (2) 39
(3) 41 (4) 43

2. The average of 25 consecutive odd integers is 55. The highest of these integers is
(1) 79 (2) 105
(3) 155 (4) 109

3. The arithmetic mean of the following numbers 1, 2, 2, 3, 3, 3, 4, 4, 4, 4, 5, 5, 5, 5, 5, 6, 6, 6, 6, 6, 6 and 7, 7, 7, 7, 7, 7, 7 is
(1) 4 (2) 5
(3) 14 (4) 20

4. The average of all the numbers between 6 and 50 which are divisible by 5 is
(1) 27.5 (2) 30
(3) 28.5 (4) 22

5. The average of 1, 3, 5, 7, 9, 11, -------- to 25 terms is
(1) 125 (2) 25
(3) 625 (4) 50

6. The average of 7 consecutive numbers is 20. The largest of these numbers is
(1) 20 (2) 23
(3) 24 (4) 26

7. The average of 8 numbers is 27. If each of the numbers is multiplied by 8, find the average of new set of numbers.
(1) 1128 (2) 938
(3) 316 (4) 216

8. The average of eight successive numbers is 6.5. The average of the smallest and the greatest numbers among them will be :
(1) 4 (2) 6.5
(3) 7.5 (4) 9

9. The average of 7 consecutive numbers is 20. The largest of these numbers is :
(1) 24 (2) 23
(3) 22 (4) 20

10. The average of first nine prime numbers is
(1) 9 (2) 11
(3) $11\frac{2}{9}$ (4) $11\frac{1}{9}$

11. The average of 5 consecutive natural numbers is m. If the next three natural numbers are also included, how much more than m will the average of these 8 numbers be?
(1) 2 (2) 1
(3) 1.4 (4) 1.5

12. The average of the first 100 positive integers is
(1) 100 (2) 51
(3) 50.5 (4) 49.5

13. The average of odd numbers upto 100 is
(1) 50.5 (2) 50
(3) 49.5 (4) 49

14. The average of the squares of first ten natural numbers is
(1) 35.5 (2) 36
(3) 37.5 (4) 38.5

15. The average of seven consecutive positive integers is 26. The smallest of these integers is :
(1) 21 (2)23
(3) 25 (4)26

16. The average of 5 consecutive natural numbers is m. If the next three natural numbers are also included, how much more than m will the average of these 8 numbers be?
(1) 2 (2) 1
(3) 1.4 (4) 1.5

17. Average of first five odd multiples of 3 is
(1) 12 (2) 16
(3) 15 (4) 21

18. The average of nine consecutive numbers is n. If the next two numbers are also included the new average will
(1) increase by 2
(2) remain the same
(3) increase by 1.5
(4) increase by 1

19. The average of four consecutive even numbers is 9. Find the largest number.
(1) 12 (2) 6
(3) 8 (4) 10

20. If a, b, c, d, e are five consecutive odd numbers, their average is
(1) $5(a + 4)$
(2) $abcde/5$
(3) $5(a + b + c + d + e)$
(4) $a + 4$

21. Average of first five prime numbers is
(1) 5.3 (2) 5.6
(3) 5 (4) 3.6

22. What is the average of the first six (positive) odd numbers each of which is divisible by 7?
(1) 42 (2) 43
(3) 47 (4) 49

23. The average of first ten prime numbers is
(1) 10.1 (2) 10
(3) 12.9 (4) 13

24. If the average of eight consecutive even numbers be 93, then the greatest number among them is
(1) 100 (2) 86
(3) 102 (4) 98

25. The average of 6 consecutive natural numbers is K. If the next two natural numbers are also included, how much more than K will the average of these 8 numbers be?
(1) 1.3 (2) 1
(3) 2 (4) 1.8

26. The average of five consecutive positive integers is n. If the next two integers are also included, the average of all these integers will
(1) increase by 1.5
(2) increase by 1
(3) remain the same
(4) increase by 2

27. The average of all the odd integers between 2 and 22 is:
(1) 14 (2) 12
(3) 13 (4) 11

28. The sum of three consecutive even numbers is 28 more than the average of these three numbers. Then the smallest of these three numbers is
(1) 6 (2) 12
(3) 14 (4) 16

29. The average of 7 consecutive numbers is 20. The largest of these numbers is
(1) 20 (2) 23
(3) 24 (4) 26

30. The average of 25 consecutive odd integers is 55. The highest of these integers is
(1) 79 (2) 105
(3) 155 (4) 10

ANSWER KEY

1.	(4)	2.	(1)	3.	(2)	4.	(1)	5.	(2)	6.	(2)	7.	(4)	8.	(2)	9.	(2)	10.	(4)
11.	(4)	12.	(3)	13.	(2)	14.	(4)	15.	(2)	16.	(4)	17.	(3)	18.	(4)	19.	(1)	20.	(4)
21.	(2)	22.	(1)	23.	(3)	24.	(1)	25.	(2)	26.	(2)	27.	(2)	28.	(2)	29.	(2)	30.	(1)

PRACTICE SHEET 2

1. 12 kg of rice costing 30 per kg is mixed with 8 kg of rice costing 40 per kg. The average per kg price of mixed rice is
(1) 38 (2) 37
(3) 35 (4) 34

2. A library has an average number of 510 visitors on Sunday and 240 on other days. The average number of visitors per day in a month of 30 days beginning with Sunday is:
(a) 285 (b) 295
(c) 300 (d) 290

3. The average weight of five persons sitting in a boat is 38 kg. The average weight of the boat and the persons sitting in the boat is 52kg. What is the weight of the boat?
(1) 228 kg (2) 122 kg
(3) 232 kg (4) 242 kg

4. The average marks of 32 boys of section A of class X is 60 whereas the average marks of 40 boys of section B of class X is 33. The average marks for both the sections combined together is
(1) 44 (2) 45
(3) $46\frac{1}{2}$ (4) $45\frac{1}{2}$

5. Six tables and twelve chairs were bought for 7,800. If the average price of a table is 750, then the average price of a chair would be
(1) 250 (2) 275
(3) 150 (4) 175

6. *a, b, c, d, e, f, g* are consecutive even numbers. *j, k, l, m, n* are consecutive odd numbers. The average of all the numbers is
(1) $3\left(\frac{a+n}{2}\right)$ (2) $\left(\frac{5l+7d}{12}\right)$
(3) $\left(\frac{a+b+m+n}{4}\right)$ (4) $\left(\frac{j+c+n+g}{4}\right)$

7. The average weight of 15 person in a boat is increased by 1.6 kg when one of the crew, who weighs 42 kg is replaced by a new man. Find the weight of the new man (in kg).
(1) 67 (2) 65
(3) 66 (4) 43

8. The average of 18 numbers is 37.5. If six numbers of average x are added to them, then the average of all the numbers increases by one. The value of x is:
(a) 42 (b) 40
(c) 38.5 (d) 41.5

9. The average of 9 numbers is 30. The average of first 5 numbers is 25 and that of the last 3 numbers is 35. What is the 6th number?
(1) 20 (2) 30
(3) 40 (4) 50

10. The average of 15 numbers is 7. If the average of the first 8 numbers be 6.5 and the average of last 8 numbers be 9.5, then the middle number is
(1) 20 (2) 21
(3) 23 (4) 18

11. Out of seven given numbers, the average of the first four numbers is 4 and that of the last four numbers is also 4. If the average of all the seven numbers is 3, fourth number is
(1) 3 (2) 4
(3) 7 (4) 11

12. The average temperature of the first 4 days of a week was 37°C and that of the last 4 days of the week was 41°C. If the average temperature of the whole week was 39°C, the temperature of the fourth day was
(1) 38°C (2) 38.5°C
(3) 39°C (4) 40°C

13. In a certain year, the average monthly income of a person was 3,400. For the first eight months of the year, his average monthly income was 3,160 and for the last five months, it was 4,120. His income in the eighth month of the year was
(1) 3,160 (2) 5,080
(3) 15,520 (4) 5,520

14. The average of 30 numbers is 12. The average of the first 20 of them is 11 and that of the next 9 is 10. The last number is
(1) 60 (2)45
(3) 40 (4)50

15. The average of 11 results is 50. If the average of the first six results is 49 and that of the last six is 52, the sixth no. is
(1) 48 (2) 50
(3) 52 (4) 56

OFFICIUM SUPREMUM

16. The average of five numbers is 27. If one number is excluded, the average becomes 25. The excluded number is :
(1) 25 (2) 27
(3) 30 (4) 35

17. The average of marks of 28 students in Mathematics was 50; 8 students left the school, then this average increased by 5. What is the average of marks obtained by the students who left the school?
(1) 50.5 (2) 37.5
(3) 42.5 (4) 45

18. The average weight of 12 parcels is 1.8 kg. Addition of another new parcel reduces the average weight by 50 g. What is the weight of the new parcel?
(1) 1.50 kg (2) 1.10 kg
(3) 1.15 kg (4) 1.01 kg.

19. The average of 50 numbers is 38. If two numbers namely 45 and 55 are discarded, the average of the remaining numbers is :
(1) 35 (2) 32.5
(3) 37.5 (4) 36

20. There are 50 students in a class. Their average weight is 45 kg. When one student leaves the class the average weight reduces by 100g. What is the weight of the student who left the class ?
(1) 45 kg (2) 47.9 kg
(3) 49.9 kg (4) 50.1 kg

21. Of the three numbers whose average is 60, the first is one fourth of the sum of the others. The first number is :
(1) 30 (2) 36
(3) 42 (4) 45

22. Of the three numbers, second is twice the first and also thrice the third. If the average of the three numbers is 44, the largest number is:
(1) 24 (2) 72
(3) 36 (4) 108

23. The average of first three numbers is thrice the fourth number. If the average of all the four numbers is 5, then find the fourth number.
(1) 4.5 (2) 5
(3) 2 (4) 4

24. Of the three numbers, first is twice the second and second is twice the third. The average of three numbers is 21. The smallest of the three numbers is
(1) 9 (2) 6
(3) 12 (4) 18

25. Of the three numbers, the first is 3 times the second and the third is 5 times the first. If the average of the three numbers is 57, the difference between the largest and the smallest number is
(1) 9 (2) 18
(3) 126 (4) 135

26. Of the three numbers, the first is twice the second and the second is 3 times the third. If their average is 100, the largest of the three numbers is :
(1) 120 (2) 150
(3) 180 (4) 300

27. Of the three numbers, the first is twice the second and the second is thrice the third. If the average of the three numbers is 10, the largest number is:
(1) 12 (2) 15
(3) 18 (4) 30

28. The average of first three numbers is double of the fourth number. If the average of all the four numbers is 12, find the 4th number.
(1) 16 (2)48/7
(3) 20 (4)18/7

29. The average of three numbers is 77. The first number is twice the second and the second number is twice the third. The first number is:
(1) 33 (2) 66
(3) 77 (4) 132

30. Out of three numbers, the first is twice the second and is half of the third. If the average of the three numbers is 56, then difference of first and third number is
(1) 12 (2) 20
(3) 24 (4) 48

ANSWER KEY																			
1.	(4)	2.	(1)	3.	(2)	4.	(2)	5.	(2)	6.	(2)	7.	(3)	8.	(4)	9.	(3)	10.	(3)
11.	(4)	12.	(3)	13.	(2)	14.	(4)	15.	(4)	16.	(4)	17.	(2)	18.	(3)	19.	(3)	20.	(3)
21.	(2)	22.	(2)	23.	(3)	24.	(1)	25.	(3)	26.	(3)	27.	(3)	28.	(2)	29.	(4)	30.	(4)

PRACTICE SHEET 3

1. The average of marks of 14 student was calculated as 71. But it was later found that the marks of one student had been wrongly entered as 42 instead of 56 and of another as 74 instead of 32. The correct average is :
(1) 67 (2) 68
(3) 69 (4) 71

2. The average weight of three men A, B and C is 84 kg. D joins them and the average weight of the four becomes 80 kg. If E whose weight is 3 kg more than that of D, replaces A, the average weight of B, C, D and E becomes 79 kg. The weight of A is
(1) 65 kg. (2) 70 kg.
(3) 75 kg. (4) 80 kg.

3. The average of a collection of 20 measurements was calculated to be 56 cm. But later it was found that a mistake had occurred in one of the measurements which was recorded as 64 cm, but should have been 61 cm. The correct average must be
(1) 53 cm (2) 54.5 cm
(3) 55.85 cm (4) 56.15 cm

4. The average of marks in Mathematics for 5 students was found to be 50. Later, it was discovered that in the case of one student the marks 48 were misread as 84. The correct average is:
(1) 40.2 (2) 40.8
(3) 42.8 (4) 48.2

5. The average weight of 15 students in a class increases by 1.5kg when one of the students weighing 40 kg is replaced by a new student. What is the weight (in kg) of the new student?
(1) 64.5 kg. (2) 56 kg.
(3) 60 kg. (4) 62.5 kg.

6. The average marks of 100 students were found to be 40. Later on it was discovered that a score of 53 was misread as 83. Find the correct average corresponding to the correct score.
(1) 38.7 (2) 39
(3) 39.7 (4) 41

7. A cricketer whose bowling average is 24.85, runs per wicket, takes 5 wickets for 52 runs and thereby decreases his average by 0.85. The number of wickets taken by him till the last match was :
(1) 64 (2) 72
(3) 80 (4) 96

8. The average of runs of a cricket player of 10 innings was 32. How many runs must he make in his next inning so as to increase his average of runs by 4?
(1) 76 (2) 70
(3) 4 (4) 2

9. The bowling average of a cricketer was 12.4. He improves his bowling average by 0.2 points when he takes 5 wickets for 26 runs in his last match. The number of wickets taken by him before the last match was
(1) 125 (2) 150
(3) 175 (4) 200

10. A cricketer had a certain average of runs for his 64 innings. In his 65th innings, he is bowled out for no score on his part. This brings down his average by 2 runs. His new average of runs is
(1) 130 (2) 128
(3) 70 (4) 68

11. A cricketer has a certain average of runs for his 8 innings. In the ninth innings, he scores 100 runs, thereby increases his average by 9 runs. His new average of runs is
(1) 20 (2) 24
(3) 28 (4) 32

12. The average of runs scored by a player in 10 innings is 50. How many runs should he score in the 11th innings so that his average is increased by 2 runs?
(1) 80 runs (2) 72 runs
(3) 60 runs (4) 54 runs

13. A cricket batsman had a certain average of runs for his 11 innings. In the 12th innings, he made a score of 90 runs and thereby his average of runs was decreased by 5. His average of runs after 12th innings is :
(1) 155 (2) 150
(3) 145 (4) 140

14. The batting average for 40 innings of a cricket player is 50 runs. His highest score exceeds his lowest score by 172 runs. If these two innings are excluded, the average of the remaining 38 innings is 48 runs. The highest score of the player is
(1) 165 runs (2) 170 runs
(3) 172 runs (4) 174 runs

15. The batting average of a cricket player for 64 innings is 62 runs. His highest score exceeds his lowest score by 180 runs. Excluding these two innings, the average of remaining innings becomes 60 runs. His highest score was
(1) 180 runs (2) 209 runs
(3) 212 runs (4) 214 runs

16. A cricket player after playing 10 tests scored 100 runs in the 11th test. As a result, the average of his runs is increased by 5. The present average of runs is
(1) 45 (2) 40
(3) 50 (4) 55

17. A cricketer has a mean score of 60 runs in 10 innings. Find out how many runs are to be scored in the eleventh innings to raise the mean score to 62?
(1) 83 (2) 82
(3) 80 (4) 81

18. In a 20 over match, the required run rate to win is 7.2. If the run rate is 6 at the end of the 15th over, the required run rate to win the match is

(1) 1.2 (2) 13.2
(3) 10.8 (4) 12

19. Average age of 6 sons of a family is 8 years. Average age of sons together with their parents is 22 years. If the father is older than the mother by 8 years, the age of mother (in years) is :
(1) 44 (2) 52
(3) 60 (4) 68

20. Out of 10 teachers of a school, one teacher retires and in his place, a new teacher of age 25 years joins. As a result, average age of teachers reduces by 3 years. The age of the retired teacher is
(1) 50 years (2) 55 years
(3) 58 years (4) 60 years

21. 3 years ago, the average age of a family of 5 members was 17 years. A baby having been born, the average age of the family is the same today. The present age of the baby is :
(1) 3 years (2) 2 years
(3) 1.5 years (4) 1 year

22. The average age of 45 persons is decreased by 1/9 year when one of them of 60 years is replaced by a new comer. Then the age of the new comer is :
(1) 45 years (2) 55 years
(3) 59 years (4) 49 years

23. When the average age of a husband and wife and their son was 42 years, the son got married and a child was born just one year after the marriage. When child turned to be five years then the average age of family became 36 years. What was the age of daughter- in-law at the time of marriage?
(1) 26 years (2) 25 years
(3) 24 years (4) 23 years

24. The average age of 30 boys in a class is 15 years. One boy aged 20 years, left the class, but two new boys came in his place whose ages differ by 5 years. If the average age of all the boys now in the class still remains 15 years, the age of the younger newcomer is :
(1) 20 years (2) 15 years
(3) 10 years (4) 8 years

25. The average age of 24 boys and their teacher is 15 years. When the teacher's age is excluded, the average age decreases by 1 year. The age of the teacher is

(1) 38 years (2) 39 years
(3) 40 years (4) 41 years

26. There were 24 students in a class. One of them, who was 18 years old, left the class and his place was filled up by a newcomer. If the average of the class thereby, was lowered by one month, the age of the newcomer is
(1) 14 years (2) 15 years
(3) 16 years (4) 17 years

27. The average age of 30 students is 9 years. If the age of their teacher is included, the average age becomes 10 years. The age of the teacher (in years) is
(1) 27 (2) 31
(3) 35 (4) 40

28. From a class of 24 boys, a boy, aged 10 years, leaves the class and in his place a new boy is admitted. As a result, the average age of the class is increased by 2 months. What is the age of the new boy?
(1) 12 years (2) 15 years
(3) 14 years (4) 13 years

29. 5 years ago, the average age of A, B, C and D was 45 years. With E joining them now, the average age of all the five is 49 years. How old is E?
(1) 25 years (2) 40 years
(3) 45 years (4) 64 years

30. In a family, the average age of a father and a mother is 35 years. The average age of the father, mother and their only son is 27 years. What is the age of the son?
(1) 12 years (2) 11 years
(3) 10.5 years (4) 10 years

ANSWER KEY

1.	(3)	2.	(3)	3.	(3)	4.	(3)	5.	(4)	6.	(3)	7.	(3)	8.	(1)	9.	(3)	10.	(2)
11.	(3)	12.	(2)	13.	(3)	14.	(4)	15.	(4)	16.	(3)	17.	(2)	18.	(3)	19.	(3)	20.	(2)
21.	(2)	22.	(2)	23.	(2)	24.	(2)	25.	(2)	26.	(3)	27.	(4)	28.	(3)	29.	(3)	30.	(2)

CDS PYQ

1. The mean of 7 observations is 7. If each observation is increased by 2, then the new mean is

[CDS 2013(I)]

(a) 12 (b) 10
(c) 9 (d) 8

2. There are 45 male and 15 female employees in an office. If the mean salary of 60 employees is Rs 4800 and the mean salary of the male employees is Rs 5000, then the mean salary of the females employees is

[CDS 2013(I)]

(a) Rs 4200 (b) Rs 4500
(c) Rs 5600 (d) Rs 6000

3. The average age of male employees is 52 years and that of female employees is 42 years the mean age of all employees is 50 years. The percentage of male and female employees is:

[CDS 2014 (I)]

(a) 80% and 20% (b) 20% and 80%
(c) 50% and 50% (d) 52% and 48%

4. If the average of A and B is 30, the average of C and D is 20, then which of the following is/are correct?
1.The average of B and C must be greater than 25.
2.The average of A and D must be less than 25.
Select the correct answer using the code given below:

[CDS 2014 (I)]

(a) 1 only (b) 2 only
(c) either 1 or 2 (d) neither 1 nor 2

5. The average of m numbers is n^4 and the average of n numbers is m^4. The average of (m + n) numbers is:

[CDS 2015 (II)]

(a) mn (b) $m^2 + n^2$
(c) $mn(m^2 + n^2)$ (d) $mn(m^2 + n^2 – mn)$

6. The average weight of students in a class is 43kg. Four new students are admitted to the class whose weights are 42kg, 36.5kg, 39kg and 42.5kg respectively. Now the average weight-of the students of the class is 42.5kg. The number of students in the beginning was:

[CDS 2015 (II)]

(a) 10 (b) 15
(c) 20 (d) 25

7. Four years ago, the average age of A and B was 18 years. Now the average age of A, B and C is 24 years. After 8 years, the age of C will be:

[CDS 2015 (II)]

(a) 32 years (b) 28 years
(c) 36 years (d) 40 years

8. The mean age of combined group of boys and girls is 18 years and the mean of age of boys is 20 and that of girls is 16, then what is the percentage of boys in the group is

[CDS 2016(II)]

(a) 60 (b) 50
(c) 45 (d) 40

9. The average score of class X is 83. The average score of class Y is 76. The average score of class Z is 85. The average score of class X and Y is 79. The average score of class Y and Z is 81.What is the average score of class X, Y and Z ?

[CDS 2016(II)]

(a) 81.5 (b) 80.5
(c) 79.0 (d) 78.0

10. A cricketer has a certain average of 10 innings. In the eleventh inning he scored 108 runs, thereby increasing his average by 6 runs. What is his new average?

[CDS 2016(II)]

(a) 42 (b) 47
(c) 48 (d) 60

11. The mean of 20 observations is 17. On checking it was found that the two observations were wrongly copied as 3 and 6. If wrong observations are replaced by correct values 8 and 9, then what is the correct mean?

[CDS 2016(II)]

(a) 17.4 (b) 16.6
(c) 15.8 (d) 14.2

12. In a class of 100 students, there are 70 boys whose average marks in a subject are 75. If the average marks of the complete class is 72, then what is the average marks of girls?

[CDS 2016(II)]

(a) 64 (b) 65
(c) 68 (d) 74

13. The mean of 5 numbers is 15, if one more number is included, the mean of 6 observations become 17. What is the included number?

[CDS 2017(I)]

(a) 24 (b) 25
(c) 26 (d) 27

14. The mean marks obtained by 300 students in a subject are 60. The mean of top 100 students was found to be 80 and the mean of last 100 students was found to be 50. The mean marks of the remaining 100 students are

[CDS 2017(I)]

(a) 70 (b) 65
(c) 60 (d) 50

15. Let a, b, c, d, e, f, g be consecutive even numbers and j, k, l, m, n be consecutive odd numbers. What is the average of all the numbers?

[CDS 2017(I)]

(a) $\frac{3(a+n)}{2}$ (b) $\frac{(5l+7d)}{12}$
(c) $\frac{a+b+m+n}{4}$ (d) None

16. A small company pays each of its 5 category 'C' workers Rs 20000, each of its 3 category 'B' workers Rs 25000 and a category 'A' workers Rs 65000. The number of workers earning less than the mean salary is

[CDS 2017(II)]

(a) 8 (b) 5
(c) 4 (d) 3

17. The average height of 22 students of a class is 140 cm, and the average height of 28 students of another class is 152cm. What is the average height of students of both the classes?

[CDS 2017(II)]

(a) 144.32 cm (b) 145.52 cm
(c) 146.72 cm (d) 147.92 cm

18. At present the average of the ages of a father and a son is 25years After 7 year the son will be 17 year old. What will be the age of the **f**ather after 10 year?

[CDS 2018(I)]

(a) 44yr (b) 45yr
(c) 50 yr (d) 52 yr

19. If the average of 9 consecutive positive integers is 55, then what is the largest integer?

[CDS 2018(I)]

(a) 57 (b) 58
(c) 59 (d) 60

20. The average of the ages of 15 students in a class is 19 year. When 5 new students are admitted to the class, the average age of the class becomes 18.5 year. What is the average age of 5 newly admitted students?

[CDS 2018(I)]

(a) 17 yr (b) 17.5 yr
(c) 18 yr (d) 18.5 yr

21. The arithmetic mean of 11 observations is 11. The arithmetic mean of the first 6 observations is 10.5 and the arithmetic mean of the last 6 observations is 11.5. What is the sixth observation?

[CDS 2018(I)]

(a) 10 (b) 10.5
(c) 11 (d) 11.5

22. The average marks of section A are 65 and that of section B are 70. If the average of both the sections combined are 67 , then the ratio of number of students of section A and to that of section B is

[CDS 2018(II)]

(a) 3 : 2 (b) 1 : 3
(c) 3 1 (d) 2 : 3

23. Let $\overline{x_1}$ and $\overline{x_2}$ (where $\overline{x_2} > \overline{x_1}$) be the means of two sets comparing n_1 and n_2 (where $n_2 < n_1$) observations respectively. If $\overline{x}$ is the mean when they are pooled. then which one of the following is correct ?

[CDS 2018(II)]

(a) $\overline{x_1} < \overline{x} < \overline{x_2}$

(b) $\overline{x} > \overline{x_2}$

(c) $\overline{x} < \overline{x_1}$

(d) $\left(\overline{x_1}-\overline{x}\right)+\left(\overline{x_2}-\overline{x_1}\right)=0$

24. In a class of 100 students, the average weight is 30 kg. If the average weight of girls is 24 kg and that of the boys is 32 kg, then what is the number of girls in class?

[CDS 2019(I)]

(a) 25 (b) 26
(c) 27 (d) 28

25. The average of 50 consecutive natural numbers is x. what will be the new average when the next four numbers also included?

[CDS 2019(I)]

(a) x + 1 (b) x + 2
(c) x + 4 (d) x + (x/54)

26. The mean of m observations out of n observations is n and the mean of remaining observations is m, then what is the mean of all n observations?

[CDS 2019(I)]

(a) $2m - (m^2/n)$ (b) $2m + (m^2/n)$
(c) $m - (m^2/n)$ (d) $m + (m^2/n)$

27. A library has an average number of 510 visitors on Sunday and 240 on other days. What is the average number of visitors per day in a month of 30 days beginning with Saturday?

[CDS 2019(II)]

(a) 276 (b) 282
(c) 285 (d) 375

28. In a class room of number of girls to that of boys is 3 : 4, The average height of students in the class is 4.6 feet. If average height of the boys in the class is 4.8feet, then what is the average height of the girls in the class?

[CDS 2020(I)]

(a) Less than 4.2 kg
(b) More than 4.5 feet and less than 4.3 feet
(c) More than 4.3 feet but less than 4.4 feet
(d) More than 4.4 feet but less than 4.5 feet

29. The ages of 7 family members are 2, 5, 12, 18, 38, 40 and 60 years respectively. After 5 years a new member aged x years is added. If the mean age of the family now goes up by 1.5 years, then what is the value of x ?

[CDS 2020(II)]

(a) 1 (b) 2
(c) 3 (d) 4

30. The mean weight of 100 students in a class is 46kg. The mean weight of boys is 50kg and that of girls is 40kg. The number of boys exceeds the number of girls by

[CDS 2020(II)]

(a) 10 (b) 15
(c) 20 (d) 25

31. The mean of five observations x, x + 2, x + 4, x + 6, x + 8 is m. What is the mean of first three observation.

[CDS 2020(II)]

(a) m (b) m - 1
(c) m - 2 (d) m - 3

32. What is the arithmetic mean of the first ten composite numbers?

[CDS 2021(I)]

(a) 8.5 (b) 9.5
(c) 10.2 (d) 11.2

33. The marks obtained by 5 students are 21, 27, 19, 26, 32. Later on 5 grace marks are added to each student. What are the average marks of the revised marks of the students?

[CDS 2021(I)]

(a) 26 (b) 30
(c) 31 (d) 32

34. Let p be the mean of m observations and q be the mean of n observations, where $p \leq q$. If the combined mean of (m + n) observations is c, then which one of the following is correct ?

[CDS 2021(I)]

(a) $c \leq p$ (b) $c \geq q$
(c) $p \leq c \leq q$ (d) $q \leq c \leq p$

35. Let the average score of a class of boys and girls in an examination be p. The ratio of boys and girls in the class is 3 : 1. If the average score of the boys is (p + 1), then what is the average score of girls?

[CDS 2021(I)]

(a) (p - 1) (b) (p - 2)
(c) (p - 3) (d) p

Direction for next two:
Collect all the sequence of five consecutive integers such that their product is equal to one of these integers. Let X be the collection of all possible such sequences. Let P be the smallest integer and Q be the largest integer occurring in these sequences.

36. How many such sequences of five consecutive integers are possible?

[CDS 2021(II)]

(a) One (b) Two
(c) Three (d) Five

37. What is the arithmetic mean of P and Q?

[CDS 2021(II)]

(a) 0

(b) 1
(c) 2
(d)Cannot be determined due to insufficient data

38. Mean marks of 50 students were found to the 78.4. But later it was detected that 95 was misread as 59 and 25 was misread as 52. What is the difference between correct mean and incorrect mean?
[CDS 2022 (II)]
(a) 0.04 (b) 0.08
(c) 0.12 (d) 0.18

39. If M is the mean of the first 100 even natural numbers, then what is the mean of the first 100 odd natural numbers?
[CDS 2022 (II)]
(a) M–1 (b) M
(c) M + 1 (d) M + 2

40. Average marks in Mathematics of Section A comprising 30 students is 65 and that of Section B comprising 35 students is 70. What are the average marks (approximately) of both the sections if it was detected later that an entry of 47 marks was wrongly made as 74?
[CDS – 2023 (1)]
(a) 67.28 (b) 67.58
(c) 68.11 (d) 68.63

41. The mean of p, q, r, s and t is 280. If the mean of p, r and t is 240, what is the mean of q and s?
[CDS-2023 (2)]
(a) 310 (b) 320
(c) 330 (d) 340

Consider the following for the next ten (10) items that follow:
Mark option (a) if the question can be answered by using one of the statements alone, but cannot be answered using the other statement alone.

Mark option (b) if the question can be answered by using either statements alone.

Mark option (c) if the question can be answered by using both the statements together, but cannot be answered using either statement alone.

Mark option (d) if the question cannot be answered even by using both the statements together.
[CDS-2023 (2)]

42. Question: Is average of the largest and the smallest of 4 given numbers greater than the average of the 4 numbers?
Statement I:
The difference between the largest and the second largest numbers is less than the difference between the second smallest and the smallest of the numbers.
Statement II:
The difference between the largest and the smallest numbers is greater than the difference between the second largest and the second smallest of the numbers.

ANSWER KEY																			
1.	c	**2.**	a	**3.**	a	**4.**	d	**5.**	d	**6.**	c	**7.**	c	**8.**	b	**9.**	a	**10.**	c
11.	a	**12.**	b	**13.**	d	**14.**	d	**15.**	b	**16.**	a	**17.**	c	**18.**	c	**19.**	c	**20.**	a
21.	c	**22.**	a	**23.**	a	**24.**	a	**25.**	b	**26.**	a	**27.**	c	**28.**	c	**29.**	b	**30.**	c
31.	c	**32.**	d	**33.**	b	**34.**	c	**35.**	c	**36.**	d	**37.**	d	**38.**	d	**39.**	a	**40.**	a
41.	d	**42.**	a																

CAPF PYQ

1. The average of 7 consecutive odd numbers is M. If the next 3 odd numbers are also included, the average

[CAPF 2017]

(a) Remains unchanged
(b) increase by 1.5
(c) increase by 2
(d) increase by 3

2. Suppose a, b, c d and e are five consecutive odd numbers in ascending order. Consider the following statements:
1. Their average is (a + 4)
2. Their average is (b + 2)
3. Their average is (e - 4)
Which of the statements given above is/are correct?

[CAPF 2018]

(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

3. The average age of the boys in a class is 12 years. The average age of the girls in the class is 11 years. There are 50% more girls than boys in the class. Which one of the following is the average age of the class (in years)?

[CAPF 2020]

(a) 11.2 years (b) 11.4 years
(c) 11.6 years (d) 11.8 years

4. The average age of raj and his father is 45 years. If the ages of the father and the grandfather of Raj are respectively two and three times that of Raj, then the age of Raj's grandfather is

[CAPF 2021]

(a) 75 years (b) 90 years
(c) 81 years (d) 84 years

5. The average age of father and elder son is 35 years, the average age of father and younger son is 32 years and the average age of the two sons is 17 years. What is the average age of the father and his sons?

[CAPF 2022]

(a) 30 years (b) 27 years
(c) 28 years (d) 29 years

6. Suppose A and B can complete a work together in 10 days. If B alone can complete the work in 15 days, then in how many days can A alone finish the work?

[CAPF 2022]

(a) 20 days (b) 24 days
(c) 25 days (d) 30 days

7. If the average of the first four of five numbers in decreasing order is 25 and the average of the last four numbers is 20, then what is the difference between the first and the last number?

[CAPF 2022]

(a) 5 (b) 10
(c) 15 (d) 20

ANSWER KEY

1.	(d)	**2.**	(d)	**3.**	(b)	**4.**	(b)	**5.**	(c)	**6.**	(d)	**7.**	(d)						